

Email forensics

Email

↳ Header || Body || Signature

Mail Recovery

Offline (PST, OST) || Online

Examining Email Messages

- ↳ Subject
- ↳ Senders address
- ↳ Email Body
- ↳ SIGNATURE
- ↳ Attachments

Challenges

- ↳ Disposable e-mail addr (Temporary)
- ↳ Anonymous emails (TOR)
- ↳ Shared email (Drafts)
- ↳ Different jurisdiction (Ruler)

★ [youtube.com/watch?v=06j6yVsc1hc](https://www.youtube.com/watch?v=06j6yVsc1hc)

Agent

MUA - User

MTA - Transfer

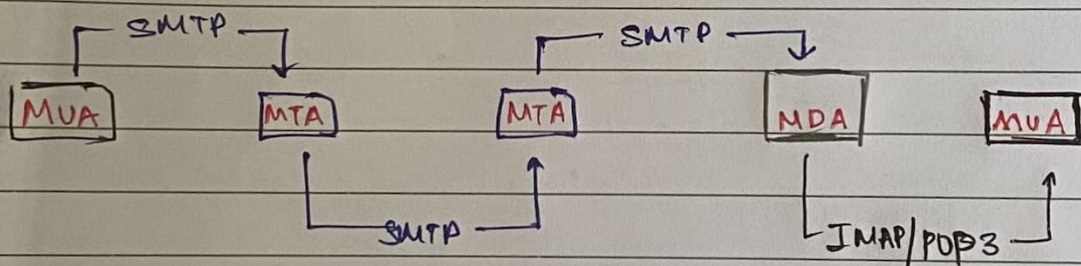
MDA - Delivery

Proto

POP3 - Post office proto 3

IMAP - iNet Msg Access proto

SMTP - Simple Mail transfer proto



Header analysis

Timestamp

From

To

Subject

MIME

messageID (FQDN)

Return Path

SPF

DKIM

DMARC

MX

MX Toolbox

+ NI PDF

Analyse email also

Email header analysis involves examining the metadata of an email to trace its origin, verify authenticity, and detect spoofing, phishing, or spam.

Key Elements:

- **From / To / Subject / Date:** Basic sender and recipient info; *easily spoofed*.
- **Return-Path:** Shows where bounce-backs go; useful for identifying spoofing.
- **Received:** Shows email's journey through servers; *read from bottom up* to find the source IP.
- **Message-ID:** Unique identifier; helps trace or group messages.
- **Reply-To:** Often manipulated in phishing.
- **SPF / DKIM / DMARC:** Authentication checks:
 - **SPF:** Is sender's IP authorized?
 - **DKIM:** Is message unchanged?
 - **DMARC:** What if  if SPF/DKIM fails?